Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Calculate the boiling point of a one molar aqueous solution (density $= 1.04\ \mathrm{gmL^{-1}}$) of potassium chloride, $\mathrm{K_b}$ for water $= 0.52\ \mathrm{kg\ mol^{-1}}$. Atomic mass of $\mathrm{K} = 39, \mathrm{Cl} = 35.5$

(A) $107.28^{\circ}\mathrm{C}$

(B) $103.68^{\circ}\mathrm{C}$

(C) $101.078^{\circ}\mathrm{C}$

(D) None of these

Q.2

The mole fraction of water in 20% aqueous solution (By weight) of $\mathrm{H_2O_2}$ is

(A) $\frac{77}{68}$

(B) $\frac{68}{77}$

(C) $\frac{20}{80}$

(D) $\frac{80}{20}$

Q.3

A weak electrolyte, AB, is 5% dissociated in aqueous solution. What is the freezing point of a 0.100 molal aqueous solution of AB ? $\mathrm{K_f}$ for water is $1.86\ \mathrm{deg/molal}$.

(A) $-3.8^{\circ}\mathrm{C}$

(B) $-0.1953^{\circ}\mathrm{C}$

(C) $-1.7^{\circ}\mathrm{C}$

(D) $-0.78^{\circ}\mathrm{C}$

Q.4

Which has maximum vapour pressure

(A) HI

(B) HBr

(C) HCl

(D) HF

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Q.5

The molal freezing point constant for water is $1.86\ \mathrm{K\ molality^{-1}}$. If $342\ \mathrm{g}$ of cane sugar $(\mathrm{C_{12}H_{22}O_{11}})$ are dissolved in $1000\ \mathrm{g}$ of water, the solution will freeze at

(A) $-1.86^{\circ}\mathrm{C}$

(B) $1.86^{\circ}\mathrm{C}$

(C) $-3.92^{\circ}\mathrm{C}$

(D) $2.42^{\circ}\mathrm{C}$

Q.6

$\mathrm{Y}\ \mathrm{g}$ of non - volatile organic substance of molecular mass M is dissolved in $250\ \mathrm{g}$ benzene. Molal elevation constant of benzene is $\mathrm{K_b}$. Elevation in its boiling point is given by

(A) $\frac{\mathrm{M}}{\mathrm{K_b Y}}$

(B) $\frac{4\ \mathrm{K_b Y}}{\mathrm{M}}$

(C) $\frac{\mathrm{K_b Y}}{4\mathrm{M}}$

(D) $\frac{\mathrm{K_b Y}}{\mathrm{M}}$

Q.7

The values of observed and calculated molecular weights of silver nitrate are 92.64 and 170 respectively. The degree of dissociation of silver nitrate is

(A) 60%

(B) 83.5%

(C) 46.7%

(D) 60.23%

Q.8

The amount of ice that will separate on cooling a solution containing $50\ \mathrm{g}$ of ethylene glycol in $200\ \mathrm{g}$ water to $-9.3^{\circ}\mathrm{C}$ is $\left[\mathrm{K_f} = 1.86\ \mathrm{K\ molality}^{-1}\right]$

(A) $38.71\ \mathrm{g}$

(B) $38.71\ \mathrm{mg}$

(C) $42\ \mathrm{g}$

(D) $42\ \mathrm{mg}$

Q.9

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The freezing point of aqueous solution contains 5% by mass urea, 1.0% by mass KCl and 10% by mass of glucose is
$\left(\mathrm{K}_{\mathrm{fH_2O}}=1.86\ \mathrm{K}\ \ \text{molality}^{\ -1}\right)$

(A) $290.2\ \mathrm{K}$

(B) $285.5\ \mathrm{K}$

(C) $269.93\ \mathrm{K}$

(D) $250\ \mathrm{K}$

Q.10

Elevation in b.p. of an aqueous urea solution is 0.52°. $\left(\mathrm{K}_{\mathrm{b}}=0.52^{\circ}\ \mathrm{mol}^{-1}\ \mathrm{kg}\ \right)$.
Hence, mole-fraction of urea in this solution is

(A) 0.982

(B) 0.0567

(C) 0.943

(D) 0.018

Q.11

Ratio of $\Delta \mathrm{T_b}/\mathrm{K_b}$ of $6\%\ \mathrm{AB_2}$ and $9\%\ \mathrm{A_2\ B}$ ($\mathrm{AB_2}$ and $\mathrm{A_2\ B}$ both are non-electrolytes) is $1\ \mathrm{mol/kg}$ in both cases. Hence, atomic masses of A and B are respectively:

(A) $60, 90$

(B) $40, 40$

(C) $40, 10$

(D) $10, 40$

Q.12

Osmotic pressure of insulin solution at $298\ \mathrm{K}$ is found to be $0.0072\ \mathrm{atm}$. Hence, height of water column due to this pressure is $[$ given $\mathrm{d(Hg)}=13.6\ \mathrm{g/mL}]$

(A) $0.76\ \mathrm{cm}$

(B) $0.70\ \mathrm{cm}$

(C) $7.4\ \mathrm{cm}$

(D) $76\ \mathrm{cm}$

Q.13

What is the osmolarity of a $0.20\ \mathrm{M}\ \mathrm{KCl}$ solution?

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(A) 0.10 osmol

(B) 0.20 osmol

(C) 0.30 osmol

(D) 0.40 osmol

Q.14

Total vapour pressure of mixture of $1\ \text{mol}$ of volatile component $\text{A}\ (P_A^o = 100\ \text{mmHg})$ and $3\ \text{mol}$ of volatile component $\text{B}(P_B^o = 60\ \ \text{mmHg}\ )$ is $75\ \text{mm}$. For such case:

(A) there is positive deviation from Raoult's law

(B) boiling point has been lowered

(C) force of attraction between A and B is smaller than that between A and A or between B and B

(D) all the above statements are correct

Q.15

Which statement is true for solution of $0.020\ \text{M}\ \text{H}_2\text{SO}_4$

(A) 2 litre of the solution contains 0.020 mole of SO_4^{2-}

(B) 2 litre of the solution contains 0.080 mole of H_3O^+

(C) 1 litre of the solution contains 0.020 mole H_3O^+

(D) None of these

Q.16

$25\ \text{mL}$ of an aqueous solution of KCl was found to require $20\ \text{mL}$ of $1\ \text{M}\ \text{AgNO}_3$ solution when titrated using a K_2CrO_4 as indicator. Depression in freezing point of KCl solution with 100% ionization will be: $\left[\text{k}_\text{f} = 2.0^\circ\text{mol}^{-1}\ \text{kg}\ \text{and}\ \mathit{molarity} = \mathit{molality}\ \right]$

(A) 5.0°

(B) 3.2°

(C) 1.6°

(D) 0.8°

Q.17

A solution containing $30\ \text{gms}$ of non-volatile solute in exactly $90\ \text{gm}$ water has a vapour pressure of $21.85\ \text{mm Hg}$ at 25°C. Further $18\ \text{gms}$ of water is then added to the solution. The resulting solution has a vapour pressure of $22.15\ \text{mm Hg}$ at 25°C. Calculate the molecular weight of the solute

(A) 74.2

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(B) 75.6

(C) 67.83

(D) 78.7

Q.18

For $0.1\ \mathrm{M}$ solution, the colligative property will follow the order

(A) $\mathrm{NaCl} > \mathrm{Na_2SO_4} > \mathrm{Na_3PO_4}$

(B) $\mathrm{NaCl} < \mathrm{Na_2SO_4} < \mathrm{Na_3PO_4}$

(C) $\mathrm{NaCl} > \mathrm{Na_2SO_4} \approx \mathrm{Na_3PO_4}$

(D) $\mathrm{NaCl} < \mathrm{Na_2SO_4} = \mathrm{Na_3PO_4}$

Q.19

The freezing point of equimolal aqueous solution will be highest for

(A) $C_6H_5NH_3^+Cl^-$ (aniline hydrochloride)

(B) $\mathrm{Ca\left(NO_3\right)_2}$

(C) $\mathrm{La\left(NO_3\right)_3}$

(D) $\mathrm{C_6H_{12}O_6}$ (glucose)

Q.20

Which one of the following statements is False

(A) The correct order of osmotic pressure for $0.01\ \mathrm{M}$ aqueous solution of each compound is $\mathrm{BaCl_2} > \mathrm{KCl} > \mathrm{CH_3COOH} >$ sucrose.

(B) The osmotic pressure (π) of a solution is given by the equation $\pi = \mathrm{MRT}$ where M is the molarity of the solution.

(C) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction.

(D) Two sucrose solutions of the same molality prepared in different solvents will have the same freezing point depression.

Q.21

Two liquids A and B form an ideal solution. when total pressure of this solution is $600\ \mathrm{torr}$ the amount fraction of A in the vapour phase 0.35 and in liquid phase is 0.70. The vapour pressure of pure A is (torr)

Q.22

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$0.004\ \mathrm{M}\ \mathrm{Na_2SO_4}$ is isotonic with $0.01\ \mathrm{M}$ glucose. Degree of dissociation of $\mathrm{Na_2SO_4}$ is ____ $(\%)$

Q.23

The vapour pressure of a solvent decreased by $10\ \mathrm{mm\ Hg}$ when a non volatile solute was added to the solvent. The mole fraction of solute in solution is 0.2, and the mole fraction of the solvent if decrease in vapour pressure is $20\ \mathrm{mm}$ of Hg, is x. The value of $5x$ is

Q.24

$20\ \mathrm{g}$ of a binary electrolyte $(\text{ mol.wt. } = 100)$ is dissolved in $500\ \mathrm{g}$ of water. The freezing point of the solution is $-0.74^{\circ}\mathrm{C}, \mathrm{K_f} = 1.86\ \mathrm{K}$. molality $^{-1}$. The degree of ionisation of the electrolyte is

Q.25

At $40^{\circ}\mathrm{C}$, the vapour pressure in torr, of methyl alcohol - ethyl alcohol solution is represented by the equation. $\mathrm{P} = 119\mathrm{X_A} + 135$ where X_A is mole fraction of methyl alcohol, then the value of $\mathrm{Lim}_{x_A \to 1} \frac{P_A}{X_A}$ is

Q.26

Persons are medically considered to have lead poisoning if they have a concentration of greater than $10\ \mathrm{micrograms}$ of lead per decilitre of blood. Concentration in parts per billion is ____g

Q.27

Moles of $\mathrm{K_2SO_4}$ to be dissolved in $12\ \mathrm{mol}$ water to lower its vapour pressure by $10\ \mathrm{mm\ Hg}$ at a temperature at which vapour pressure of pure water is $50\ \mathrm{mm}$ is

Q.28

When $20\ \mathrm{g}$ of naphthoic acid $(\mathrm{C_{11}H_8O_2})$ is dissolved in $50\ \mathrm{g}$ of benzene $(\mathrm{K_f} = 1.72\ \mathrm{K\ kg\ mol^{-1}}$), a freezing point depression of $2\ \mathrm{K}$ is observed. The van't Hoff factor $(i) \times 10$ is

Q.29

The van't Hoff factor of the compound $\mathrm{K_3Fe(CN)_6}$ is

Q.30

The mole fraction of a solute in carbon tetrachloride is 0.235. The molality of the solution is about ____ $\mathrm{mol/kg}$

Answers & Solutions

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Q.1 Answer:

$101.078^{\circ}\mathrm{C}$

Solution:

Volume of solution $= 1000\ \mathrm{ml}$

Mass of the solution $= \mathrm{V} \times \mathrm{d}$

$= 1000\ \mathrm{mL} \times 1.04\ \mathrm{g/mL} = 1040\ \mathrm{g}$

Amount of solute in $1000\ \mathrm{mL}$ **solution**

$= 1\ \mathrm{M}$

$= 1\ \mathrm{M} \times$ **Molecular Mass** $= 1\mathrm{M} \times 74.5\ \mathrm{g/mol}$

$= 74.5\ \mathrm{g}$

Mass of water = Mass of solution − Mass of

$\mathrm{KCl} = 1040\ \mathrm{g} - 74.5\ \mathrm{g} = 965.5\ \mathrm{g}$

Molality of the solution $= \mathrm{m} = \frac{\text{Moles of solute}}{\text{Mass of solvent in kg}} = \frac{1\ \mathrm{mol}}{(965.5/1000)\mathrm{kg}}$

$= 1.0357$

$\Delta \mathrm{T_b} \mathrm{i}.\mathrm{K_b} \cdot \mathrm{m} = 2 \times 0.52\ \mathrm{K\ kg\ mol^{-1}} \times 1.0357$

$= 1.078^{\circ}\mathrm{C}.$

Boiling point of the solution

$= 100^{\circ}\mathrm{C} + 1.078^{\circ}\mathrm{C} = 101.078^{\circ}\mathrm{C}$

Q.2 Answer:

$\frac{68}{77}$

Solution:

Mole fraction of $\mathrm{H_2O} = \frac{\frac{80}{18}}{\frac{80}{18} + \frac{20}{34}} = \frac{68}{77}$

Q.3 Answer:

$-0.1953^{\circ}\mathrm{C}$

Solution:

Degree of dissociation, α, $AB = \frac{5}{100} = 0.05$

No. of moles dissolved

	AB	$\rightarrow \mathrm{A^+} + \mathrm{B^-}$	
	m	0	0
No. of moles after dissociation	$\mathrm{m}(1-\alpha)$	$\mathrm{m}\alpha$	$\mathrm{m}\alpha$
After dissociation	$0.1(1-0.05)$	0.1×0.05	0.1×0.05

Total moles = Molality

$= 0.1(1 - 0.05) + 0.1 \times 0.05 + 0.1 \times 0.05$

$= 0.095 + 0.005 + 0.005 = 0.105\ \mathrm{m}$

$\Delta \mathrm{T_f} = \mathrm{K_f m}$ **or** $\mathrm{T_f^0} - \mathrm{T_f} = 1.86\ \mathrm{K/m} \times 0.105\ \mathrm{m}$

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$= 0.1953$ deg
$T_f = 0^\circ C - 0.1953 = -0.1953^\circ C$

Q.4 Answer:

HCl

Solution:

The lower is boiling point more is vapour pressure; boiling point order,
$HCl < HBr < HI < HF$
HCl is the most volatile compound among all hydrogen halides. The volatility of a compound is related to its high vapour pressure at ordinary temperature. HF due to hydrogen bonding is least volatile, as the molecular weight increases, intermolecular forces increases due to van der Waals' force of attraction. Thus, HI is liquid and HCl is gas with maximum vapour pressure among all hydrogen halides.

Q.5 Answer:

$-1.86^\circ C$

Solution:

$\Delta T_f = \frac{1000 \times K_f \times w}{Mw \times W}$
$= \frac{1000 \times 1.86 \times 342}{342 \times 1000} = 1.86$ K
$\therefore T_f = 0 - 1.86 = -1.86^\circ C$

Q.6 Answer:

$\frac{4\,K_b Y}{M}$

Solution:

$\Delta T = \frac{1000 \times K_b \times Y}{250 \times M} = \frac{4\,K_b Y}{M}$

Q.7 Answer:

83.5%

Solution:

i for $AgNO_3 = \frac{\text{Normal mol.wt}}{\text{Observed mol.wt.}} = 1 + \alpha$
$\alpha = \frac{170}{92.64} - 1 = 0.835 = 83.5\%$

Q.8 Answer:

38.71 g

Solution:

$\Delta T = \frac{1000 \times K_f \times w}{W \times M}$
or $9.3 = \frac{1000 \times 1.86 \times 50}{62 \times W}$

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$\therefore \mathrm{W} = 161.29$

$\therefore$ **Ice separated** $= 200 - 161.29 = 38.71$ g

Q.9 Answer:

269.93 K

Solution:

$\Delta \mathrm{T_f} = \Delta \mathrm{T_f}$ **for glucose** $+ \Delta \mathrm{T_f}$ **for** $\mathrm{KCl} + \Delta \mathrm{T_f}$ **for urea**

$= \frac{1000 \times 1.86 \times 10}{100 \times 180} + \frac{1000 \times 1.86 \times 1 \times 2}{74.5 \times 100} + \frac{1000 \times 1.86 \times 5}{100 \times 60} = 3.069^{\circ}$

$\therefore$ **freezing point** $= 273 - 3.069 = 269.93$ K

Q.10 Answer:

0.018

Solution:

$\Delta \mathrm{T_b} =$ **molality** $\times \mathrm{K_b}$

$0.52 = m \times 0.52$

molality $= 1 \text{ mol kg}^{-1}$

$\therefore$ **urea** $= 1$ mol

water $= \frac{1000}{18} = 55.55$

mole fraction of urea $\frac{1}{55.55} = 0.018$

Q.11 Answer:

40, 10

Solution:

$\Delta T = \frac{1000 K_b w_1}{m_1 w_2}$

$\frac{\Delta T}{K_b}\left(AB_2\right) = \frac{1000 \times 6}{m_1 w_2} = 1$

$\frac{1000 \times 6}{m_1 \times 100} = 1$

$\therefore m_1\left(AB_2\right) = 60 = A + 2B$

$\frac{\Delta T}{K_b}\left(A_2B\right) = \frac{1000 \times 9}{m_1 \times 100} = 1$

$\therefore m_1\left(A_2B\right) = 90 = 2A + B$

$\therefore \mathrm{A} = 40, \ \mathrm{B} = 10$

Q.12 Answer:

7.4 cm

Solution:

$\mathrm{P} = 0.0072$ atm

$= 0.0072 \times 76$ cm **of** Hg

$= 0.0072 \times 76 \times 13.6$ cm **of mercury column**

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$= h \times 1$ **of water column**
$h = 0.0072 \times 76 \times 13.6$ cm
$= 7.4$ cm

Q.13 Answer:

0.40 osmol

Solution:

$KCl \rightleftharpoons K^+ + Cl^-$

Osmosis is a colligative property and thus

$0.2\text{M} = 0.4\text{M}$

Q.14 Answer:

all the above statements are correct

Solution:

$P = P_A^o X_A + P_B^o X_B$
$= \frac{100}{4} + \frac{60 \times 3}{4}$
$= 70$ mm < 75 mm **(experimental)**

Thus, there is positive deviation (a) true mixture is more volatile due to decrease in b.p. Thus, (b) is true also force of attraction is decreased thus (c) is true.

Q.15 Answer:

2 litre of the solution contains 0.080 mole of H_3O^+

Solution:

$[\text{H}_3\text{O}^+] = 2 \times 0.02 = 0.04\text{M}$
$\therefore$ **2 litre solution contains** 0.08 **mole of** H_3O^+

Q.16 Answer:

3.2°

Solution:

$M(KCl) = \frac{20 \times 1}{25} = 0.8\text{M}$
$i = [1 + (y - 1)x] = 2$
$\Delta\text{T}_\text{f} =$ **molality** $\times \text{K}_\text{f} \times \text{i}$
$= 0.8 \times 2 \times 2 = 3.2^\circ$

Q.17 Answer:

67.83

Solution:

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We have,

$\frac{p^{\circ}-21.85}{21.85}=\frac{30\times18}{90\times m}$ **for I case.....(i)**

wt. of solvent $=90+18=108$ **gm**

$\frac{p^{\circ}-22.15}{22.15}=\frac{30\times18}{108\times m}$**, for II case.....(ii)**

By eq. (1)

$p^{\circ}m-21.85m=21.85\times6=131.1$

By eq. (2)

$p^{\circ}m-22.15m=22.15\times5=110.75$

$0.30m=20.35$

$m=\frac{20.35}{0.30}=67.83$

Q.18 Answer:

$\mathrm{NaCl}<\mathrm{Na_2SO_4}<\mathrm{Na_3PO_4}$

Solution:

Colligative property in decreasing order

$\mathrm{Na_3PO_4}>\mathrm{Na_2SO_4}>\mathrm{NaCl}$

$\mathrm{Na_3PO_4}\rightarrow3\mathrm{Na^+}+\mathrm{PO_4^{3-}}=4$

$\mathrm{Na_2SO_4}\rightarrow2\mathrm{Na^+}+SO_4^{2-}=3$

$\mathrm{NaCl}\rightarrow\mathrm{Na^+}+\mathrm{Cl^-}=2$

Q.19 Answer:

$\mathrm{C_6H_{12}O_6}$ (glucose)

Solution:

$\mathrm{La\,(NO_3)_3}$ **will furnish four ions and thus will develop more lowering in freezing point whereas glucose gives only one particle and thus minimum lowering in freezing point.**

Q.20 Answer:

Two sucrose solutions of the same molality prepared in different solvents will have the same freezing point depression.

Solution:

The extent of depression in freezing point varies with the number of solute particles for a fixed solvent only and it is a characteristic feature of the nature of solvent also. So for two different solvents the extent of depression may vary even if the number of solute particles is dissolved.

Q.21 Answer:

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Solution:

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$P_{\text{total}} = 600$ **torr** $P_A = P \times X_A \ldots\ldots$ **(i)**

$(\mathrm{P} =$ **total pressure** $)$

$\mathrm{X_A} =$ **mole fraction of** A **in vapour pressure**

$\mathrm{Y_A} =$ **mole fraction of** A **in liquid solution**

$\mathrm{P_A} = \mathrm{P_A^0} \times \mathrm{Y_A} \ldots\ldots$ **(ii)**

$\mathrm{P_A} =$ **vapour pressure of pure** A

from Eq. (1) **and** (2) $\mathrm{P_A^0} \times \mathrm{Y_A} = \mathrm{P_{total}} \times \mathrm{X_A}$

$\mathrm{P_A^0} \times 0.70 = 600 \times 0.35$

$\mathrm{P_A^0} = 600 \times \frac{0.35}{0.70} = \frac{600}{2} = 300$ torr

Q.22 Answer:

—

Solution:

0.004 M $\mathrm{Na_2SO_4}$ **solution is isotonic with**

0.01 M **solution of glucose, so their osmotic pressures are equal to each other.**

Osmotic pressure of 0.01 M **glucose**

$(\pi)_{\text{glucose}} = CST$

$\therefore \mathrm{C} =$ **Concentration of solution** $= 0.01$ M

$\mathrm{S} =$ **Solution constant** $= 0.0821$ L $-$ atm/K/mol

$\mathrm{T} =$ **Absolute temperature**

$\therefore \pi_{\text{glucose}} = 0.01 \times 0.0821 \times \mathrm{T} \ldots\ldots$ **(i)**

$\pi_{\text{glucose}} = \pi_{Na_2SO_4}$

$\mathrm{Na_2SO_4}$ **is present in ionic state in solution**

So, $\mathrm{Na_2SO_4} \rightleftharpoons 2\mathrm{Na^+} + \mathrm{SO_4}^{2-}$

$\quad \mathrm{At\ t} = 0 \qquad 1 \qquad 0 \qquad 0$

At equilibrium $1-\alpha \quad 2\alpha \qquad \alpha$

(where α **is the degree of dissociation of** $\mathrm{Na_2SO_4}$**)**

By van't Hoff factor

$\frac{(\pi_{obs})_{Na_2SO_4}}{(\pi_{cal})_{Na_2SO_4}}$

$= \frac{\text{number of particles after dissociation}}{\text{number of particles before dissociation}}$

$= \frac{1-\alpha+2\alpha+\alpha}{1}$

$\because (\pi_{obs})_{\mathrm{Na_2SO_4}} = \pi_{glucose}$

$\therefore \frac{0.01 \times 0.0821 \times T}{0.004 \times 0.0821 \times T} = \frac{1+2\alpha}{1}$

or $\frac{10}{4} = \frac{1+2\alpha}{1}$ **or** $10 = 4 + 8\alpha$

$\alpha = \frac{10-4}{8} = 0.75$

Percentage of $\alpha = 75\%$

Q.23 Answer:

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Solution:

$\mathrm{P}^{\circ}-\mathrm{P}_{\mathrm{s}}=\mathrm{P}^{o} \times$ **mole fraction of solute**
$10=\mathrm{P}^{o} \times 0.2$
Again, $20=\mathrm{P}^{o} \times$ **mole fraction of solute**
Hence, $\mathrm{n}=0.4$**, so, mole fraction of solvent**
$=1-0.4=0.6$

Q.24 Answer:

—

Solution:

$\Delta \mathrm{T}_{\mathrm{f}}=\frac{1000 \times \mathrm{K}_{\mathrm{f}} \times \mathrm{w}}{\mathrm{M} \times \mathrm{W}}$
$\therefore 0.74=\frac{1000 \times 1.86 \times 20}{\mathrm{M} \times 500} \Rightarrow \mathrm{M}=100$
Now, $\frac{\text{Normal mol.wt}}{\text{exp. mol.wt.}}=1+\alpha=\frac{100}{100}$
$\mathrm{So}, \alpha=0$

Q.25 Answer:

—

Solution:

$\mathrm{P}=119 \mathrm{X}_{\mathrm{A}}+135 ; \ \mathrm{Lim}_{\mathrm{x}_{\mathrm{A}} \rightarrow 1} \frac{\mathrm{P}_{\mathrm{A}}}{\mathrm{X}_{\mathrm{A}}}$
$=119+135=254$

Q.26 Answer:

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Solution:

$10 \mu \mathrm{g}$ **per decilitre**
$\Rightarrow 10 \times 10^{-6}$ g **in** 100 mL
$\therefore 10^{9}$ **parts (** 1 **billion) has**
$=\frac{10 \times 10^{-6}}{100} \times 10^{9} g=100 g$

Q.27 Answer:

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Solution:

Lowering of V.P. is colligative property
thus, $\mathrm{i}\left(\mathrm{K}_{2} \mathrm{SO}_{4}\right)=1+(\mathrm{y}-1) x=1+2 x=3$
$\therefore$ **If** $\frac{\Delta p}{p^{\circ}}=\frac{n_{1} i}{n_{1} i+n_{2}}$
$\frac{10}{50}=\frac{3 n_{1}}{3 n_{1}+12}=\frac{n_{1}}{n_{1}+4}$

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Q.28 Answer:

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Solution:

Molecular weight of naphthoic acid, $C_{11}H_8O_2$ **is** $172\ g\ mol^{-1}$

The theoretical value of depression in freezing point

$= k_f \times$ **molality** $= 1.72 \times \frac{20 \times 1000}{172 \times 50} = 2\ K$

$\therefore$ **Van't Hoff factor i** $= \frac{\text{observed value of colligative property}}{\text{Theoretical value of colligative property}}$

$= \frac{2}{4} = 0.5$

Q.29 Answer:

—

Solution:

$K_3\left[Fe(CN)_6\right] \rightarrow 3K^+ + \left[Fe(CN)_6\right]^{3-}$

The Vant Hoff factor (i) for a dilute solution of $K_3[Fe(CN)_6]$ **is 4 . It dissociates to give 3 potassium ions and** $1\ \left[Fe(CN)_6\right]^{3-}$ **ion.**

Q.30 Answer:

—

Solution:

Let

Amount of solute $= 0.235\ mol;$

Amount of solvent $= (1 - 0.235) mol = 0.765\ mol$

Mass of solvent,

$m = nM_m = (0.765\ mol)\left(154\ g\ mol^{-1}\right) = 117.81\ g$

Molality of solution $= \frac{n}{m} = \frac{0.235\ mol}{117.81 \times 10^{-3}\ kg} \simeq 2.0\ mol\ kg^{-1}$

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